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Project work

Bayesian Optimisation with Differentiable Simulated Priors for Particle Accelerator Tuning

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Project Work **P — XX XX-XX —**
for Ms. Hiba Sikander Student ID: 610416 Study Course: L2353

Title:
Bayesian Optimisation with Differentiable Simulated Priors for Particle Accelerator Tuning

Project Description:
The tuning of particle accelerators is a time-consuming and difficult task that continues to be performed manually in many cases, reducing the scientific output of some of the most advanced but also costly research facilities in the world. In the field of autonomous accelerator tuning, researchers aim to develop methods that assist human operators and thereby reduce tuning times and extend the experimental capabilities of accelerators. One of the most promising approaches to solve the autonomous accelerator tuning problem is the use of Bayesian Optimisation (BO), which is already successfully used in operations. [1]

However, the most interesting accelerator tuning tasks are characterised by their high dimensionality. Despite being more capable at dealing with many dimensions than other algorithms, BO struggles to solve optimisations problems with more than a few dozen dimensions. To solve this challenge, researchers have proposed to leverage domain knowledge by replacing the commonly used but uninformative zero mean prior with a model of the system under optimisation. So far neural networks have been used for this purpose as they are differentiable and fast to compute. [2] While this work has shown the advantages of using an informed neural network prior, neural networks come with the disadvantage that they require large amounts of data to be trained, and that outside of the training data domain, their behaviour is ill-defined. In recent developments, high-speed differentiable simulators such as Cheetah [3] have been proposed as an alternative for use as a prior in BO, with the key advantage that they do not require any training data, and are well-behaved throughout. [3]

The goal of this research project is to take the BO with Cheetah prior approach presented on a very simple example in [3], and implement it on the closer-to-real world as well as more complex transverse beam parameter tuning task at the ARES accelerator, which has already been extensively studied with other autonomous tuning methods. The developed implementation should then be evaluated according to previous evaluations, to establish how this improved variation of BO compares to other methods and whether extension to even more complex accelerator tuning tasks is a promising avenue for future research.

Tasks:

1. Implement BO with Cheetah prior for tuning the transverse beam parameters on a simulation of the ARES Experimental Area.
2. Evaluate the BO with Cheetah prior implementation in accordance with the evaluation in [4].
3. Evaluate timing and speed of BO with an uninformed prior vs. BO with a Cheetah informed prior.
4. Study performance under model uncertainties.
5. **(Optional depending on accelerator availability)** Evaluate the developed implementation on the real ARES accelerator in accordance with [4].
6. **(Optional if time allows)** Setup a general implementation of BO with Cheetah.

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[3] J. Kaiser, C. Xu, A. Eichler, and A. Santamaria Garcia, “Bridging the gap between machine learning and particle accelerator physics with high-speed, differential simulations”, *Physical Review Accelerators and Beams*, 2024.

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Deutscher Titel:

Start Date: 17.11.2025
Due Date: 02.03.2026

February 11, 2026, Prof. Dr.-Ing Annika Eichler

Hereby I declare that I produced the present work myself only with the help of the indicated aids and sources.

Hamburg, February 11, 2026 Hiba Sikander

Abstract

Tuning particle accelerators is an expensive and time consuming challenge that causes limitations in the findings of advanced research facilities. Bayesian Optimization (BO) has shown potential to be a promising approach for autonomous tuning of accelerators, yet its effectiveness diminishes for higher dimensional problems. Recent studies have proposed replacing the zero mean prior in BO with physics informed prior models to leverage physics domain knowledge. The Cheetah framework, a high-speed differentiable simulator, was demonstrated as a physics informed BO prior on a simple two dimensional FODO lattice. This project work extends that approach to a real world based five dimensional problem now, which is the transverse beam parameter optimization at ARES Experimental Area involving three quadrupole magnets and two corrector magnets. This project implements a framework that incorporates Cheetah as the Gaussian Process mean function into the Xopt optimization toolset with trainable misalignment parameters that enable the prior to adjust when the simulator does not accurately represent the real system. Four optimization approaches are evaluated. Results from 10 trails show that physics informed BO performed significantly better than baseline techniques. Both Cheetah prior variants (matched and mismatched prior) achieve 98.4% improvement in beam error with mean absolute error of 0.006 mm, compared to standard BO with 88.9% and 0.046 mm, and Nelder-Mead with 83.7% and 0.067 mm. While standard BO only reaches 20% and Nelder-Mead 0%, both Cheetah variants managed to achieve a 100% success rate in achieving the desired threshold (40 μm). Moreover, the matched prior converged the fastest, requiring only 4 steps (median) compared to 14 for mismatched prior. These results validate the scalability of differentiable simulator priors simple and straightforward to realistic and complex accelerator tuning problems.

Contents

1 Introduction

1.1 Motivation

Particle accelerators are one of the most advanced scientific devices ever developed to perform innovative scientific experiments. The Deutsches Elektronen-Synchrotron (DESY) [1] located in Hamburg region hosts multiple top notch accelerator facilities such as European XFEL X-ray Laser and the PETRA III. These facilities accelerate electrons to almost the speed of light, producing intense beams of X-rays that enable researchers to study atomic level processes and structures with great precision. A crucial challenge for particle accelerators is the tuning process, which involves adjustment of many magnetic components to produce desired beam properties including position, size and divergence. As a specialized test facility for accelerators research and development, the ARES (Accelerator Research Experiment at SINBAD) linear accelerator at DESY allows the evaluation of innovative tuning techniques prior to deployment on even larger facilities.

Beam tuning is essentially an optimization task. In order to find exact configurations that minimize the deviation between the actual beam properties and the desired beam, we optimize the adjustable parameters (like magnet strengths and corrector/steering angles). However, the optimization is not that simple and perhaps contains a number of challenges:

1. **Expensive Evaluation:** It is impractical to conduct an extensive search since measurement of each beam property takes time and resources.
2. **Uncertainity:** There are a number of systematic errors and noises that might affect beam measurement.
3. **Black-box Nature:** The complex relationship between input parameters and beam characteristics cannot be directly observed analytically during operation.
4. **Unknown / Partial Observability:** Important parameters like characteristics of incoming beam and magnet misalignments are frequently either unknown or partially observed.

A traditional approach to accelerator tuning could be through skilled professionals who manually adjust parameters using their knowledge and experience. While being efficient, this method significantly relies on individual skill, takes a lot of time, is expensive and does not scale well to more complicated accelerator systems. Therefore, automated optimization methods are necessary for effective accelerator performance. Several optimization algorithms have been applied to particle accelerator tuning problems with their own unique advantages and disadvantages. Nelder-Mead Simplex is a gradient free optimization technique and one of the approaches that is widely used because of its simplicity and robustness [2]. However they do not create models that generalize across many optimization runs and usually need many function evaluations to converge. Another approach is the Reinforcement Learning (RL) technique that has shown remarkable performance [3]. RL agents can learn to tune accelerators in very few steps during deployment by training neural network regulations in simulation. However, RL requires extensive training data as well as very careful reward function design. The trained policies could not generalize well to conditions outside of their training distribution. The approach of our interest for this project work is Bayesian Optimization (BO), which finds a middle ground among these approaches. BO creates a probabilistic surrogate model of the objective function, which is usually a Gaussian Process, that provides estimates of uncertainty as well as predictions. This makes it possible for the assessment points to wisely balance between both exploration and exploitation. BO is a desirable option for accelerator tuning since it excels at solving expensive black-box optimization problems.

Another key insight motivating this project work is that, since the physics of particle accelerators is already well understood now, We can map the passage of charged particles via magnets using a reliable model and then they travel through magnetic fields according to well established physics knowledge. Thus, all thanks to modern simulation library like Cheetah [4] that we can now predict beam behavior with high accuracy. The standard BO uses uninformed zero mean prior function in the Gaussian process model. This implies that the algorithm must learn the whole function landscape from scratch and have to begin with no prior knowledge of the intended objective function values. On the other hand, if the physics informed knowledge could be incorporated into the prior, this shall give advantage and optimization could already start with meaningful predictions that could focus its limited evaluation budget on identifying the differences between the model and the reality. The importance of physics informed prior for accelerator optimization has also been shown in the recent work [5] where the neural network surrogate models trained on the simulation data can perform as effective prior mean functions, significantly speeding up the convergence. However, training neural network surrogates requires very large datasets from simulations. So let's continue studying Bayesian Optimization with Cheetah informed prior.

1.2 Extension from Simple FODO to Complex ARES Problem

Cheetah has already been demonstrated as a physics informed prior for BO on a straightforward beam focusing simple FODO (Focus-Drift-Defocus-Drift) lattice [5]. A FODO cell is a fundamental building block made up of alternating quadrupole magnets for focusing and defocusing that are separated by drift spaces.

This optimizational problem was two dimensional and only involved strengths of two quadrupole magnets (k^{Q1} and k^{Q2}) with the objective of minimising beam size at target location, defined as $0.5(|\sigma_x| + |\sigma_y|)$. One trainable hyperparameter, the drift length between the quadrupoles, which was restricted to the interval [0.2, 1.0] m, was included in the physics informed prior mean function. The results showed that BO with Cheetah prior converged about 15 steps, indicating better sample efficiency than Nelder Mead simplex and standard BO with zero prior. Furthermore, the method worked even with a mismatched prior, the algorithm learned the correct values during GP model fitting and converged to the true optimum when the drift lengths in the prior model (0.5m) was different from the ground truth (0.7m).

While this validated the fundamental concept, the FODO problem is an oversimplified example which does not depict the complex nature of actual accelerator tuning operations. The two dimensional search space is quite easy to explore and the model reality mismatch was restricted to a single geometric parameter (drift length) instead of the various other sources of uncertainty found in actual accelerators.

This project extends the FODO proof of concept to a significantly more complex and realistic problem which is transverse beam tuning at the ARES experimental area at DESY. The extension contain following key advancements:

1. **Higher Dimensionality:** In contrast to two dimensional FODO problem, ARES problem possess five dimensional search space including two steering/corrector magnet angles (θ_{CV} and θ_{CH}) and quadrupole strength (k^{Q1}, k^{Q2}, k^{Q3}). The volume of the search space is greatly increased by this 2.5x increase in dimensionality, which makes more difficult and raises the importance of informed prior.

2. **Realistic Lattice:** the ARES experimental area ($ARESlatticeStage3v1_9.json$) is a real accelerator section loaded from an actual lattice description file unlike the synthetic FODO cell.

3. **Objective Nature:** for Ares problem, the objective function combines four beam parameters into a single metric: $0.25(|\mu_x| + |\sigma_x| + |\mu_y| + |\sigma_y|)$ including horizontal and vertical positions and sizes. This makes simultaneous optimization of beam centering(position) and focusing(size).

4. **Trainable Misalignment Parameters:** Under this project work, the use of quadrupole misalignments as the cause of model reality mismatch is an important step forward. The ARES prior includes 6 trainable misalignment parameters (horizontal and vertical for each quadrupoles Q1, Q2 and Q3) which are limited to physically plausible intervals of $\pm 0.5mm$, in contrast to the FODO demonstration, where only a single drift length was learned.

1.3 Contribution

Thus the main contributions of this work are :

1. Demonstrates that Cheetah informed prior approach is scalable from 2D synthetic lattice to 5D real world accelerator tuning problem.

2. Implements physics informed prior mean function for the ARES lattice being compatible with the Xopt Bayesian optimisation framework

3. Extends prior with six trainable misalignment parameters, enabling the physics model to adapt to real accelerator conditions during optimization

4. Contains a comprehensive comparison between standard BO, Nealder-Mead and physic-informed BO (matched and mismatched prior) across multiple trials.

2 Theory

In this chapter, the theoretical foundations of this work are developed. We begin with Gaussian processes as probabilistic function approximators, present Bayesian optimization as a framework for sample efficient black-box optimization and then eventually look over the Cheetah simulator and the ARES accelerator where the work is used and how physics informed knowledge may be integrated through the prior mean function.

2.1 Gaussian Process

Gaussian process (GP) offers a robust framework for regression which allows uncertainty quantification along with predictions. GPs are well suited for modeling unknown objectives in optimization problems [6] because they define probability distributions directly over functions, unlike parametric models

+that learn a predefined set of weights. A Gaussian process is fully specified by its mean function $m(x)$ and covariance function (kernel) $k(x, x')$ that encodes assumptions about the function's smoothness:

$$f(x) \sim \mathcal{GP}(m(x), k(x, x')) \tag{2.1}$$

The kernel carries assumptions regarding smoothness and correlation between points, the mean function encodes previous expectations regarding the behavior of the function. The kernel selection determines what kind of functions the GP considers probable. Because of its ability in modeling smooth, physically realistic functions without assuming infinite differentiability, Matérn-5/2 kernel is widely used in Bayesian Optimization [6]. The GP uses Bayes's rule (formula for baye's rule) to update from prior to posterior when we observe data. The posterior mean remains Gaussian, with mean and variance that reflects both our prior assumption and observed evidence. One of the important characteristics is that posterior uncertainty is low in areas with dense observations and high in areas with sparse data, this natural uncertainty quantification is what makes GPs useful for guiding optimization.

2.2 Bayesian Optimization

Bayesian Optimization (BO) is a sequential method for determining the optimal value of expensive black-box functions [6]. It is especially effective when gradient information is not available and each function evaluation is expensive. Because of these characteristics, BO is ideally suited for accelerator tuning, where it has emerged as a cutting-edge method [7] [8]. The goal is to solve:

$$x^* = \arg\min_{x \in \mathcal{X}} f(x) \tag{2.2}$$

Here, x represents the control variables (like magnet settings), $\mathcal{X}$ is the search space and $f(x)$ is the quantity to be minimised (beam error in our case). BO aims to find x^* using as few function evaluation as possible. The key idea is to use a probabilistic surrogate model (usually a GP) to guide the search rather than evaluating the expensive function everywhere.

function. Exploration (sampling when uncertainty is large) and exploitation (sampling where the predicted value is good) are balanced by the acquisition function. After evaluating the true objective at the selected point, we update the dataset and repeat until the evaluation budget is exhausted.

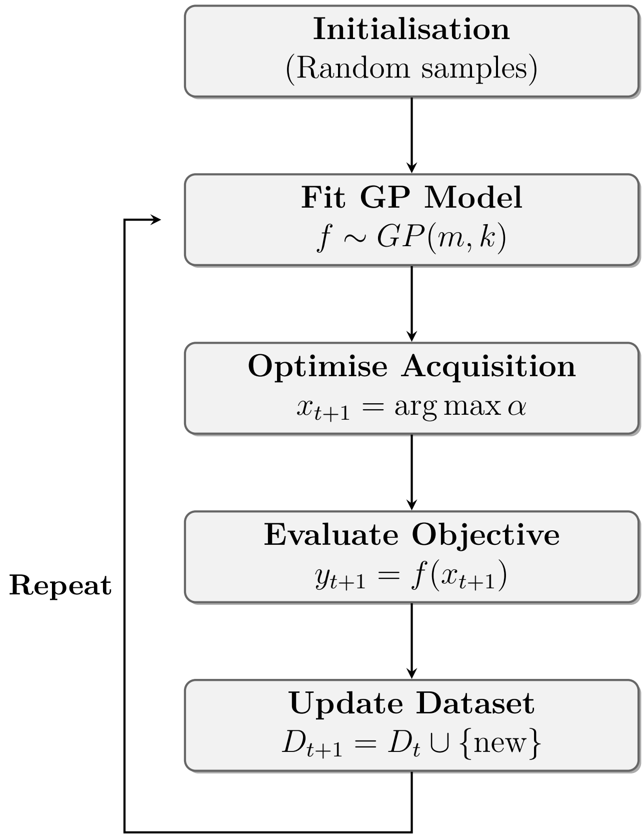

Figure 2.1: The Bayesian optimisation loop.

In this project work, the predicted mean and uncertainty are directly combined using the Upper Confidence Bound (UCB) acquisition function:

$$\alpha_{UCB}(x) = \mu(x) + \beta\sigma(x) \tag{2.3}$$

where $\mu(x)$ and $\sigma(x)$ are the GP posterior mean and standard deviation respectively. The parameter β controls the exploration-exploitation trade-off. For minimization, this formulation prefers points that either have low predicted values (exploitation) or high uncertainty (exploration). We use $\beta = 2.0$ throughout this project work, following the common practice [6] .

2.3 Physics Informed Prior Mean Function

Before any data is observed, our belief about the objective is represented by the prior mean function $\mu(x)$. The standard practice is to use zero or constant mean, which assumes no prior knowledge. However, physics models that approximate the system are often used in scientific applications, incorporating this knowledge can significantly improve the performance [5] .

To understand why, consider the GP posterior mean prediction at a new point. The GP learns to predict the difference between the prior mean and the actual function. These residuals are small and can be learned from fewer observations when the prior mean is accurate. Whereas when the prior is simply zero, the GP has to learn the entire function from scratch. This motivates using a physics informed similar as the prior mean:

$$m(x) = f_{physics}(x; \theta) \tag{2.4}$$

Where θ represents the model parameters like element alignments or beam characteristics. This benefits to improve sample efficiency, have meaningful predictions before observing data and physical consistency in extrapolation.

A key innovation in this work is making the physics model parameters θ trainable alongside the standard GP hyperparameters. In standard GP practice, the kernel hyperparameters ω (length scales, signal variance) and noise variance are estimated by maximising the log marginal likelihood of observed data [9] . As theeta influences the prior mean predictions, it can be included in this optimisation.

$$\theta^* = \arg\max_{\theta} \log p(y|X, \theta, \omega) \tag{2.5}$$

For this optimization to work, the physics similar must be differentiable so that we can compute how changes in the model parameters θ affect the marginal likelihood. Cheetah satisfies this requirement through its PyTorch implementation. As the optimization progresses and more data is collected, the GP fitting process automatically adjusts the physics model parameters to better match the observed beam behavior. This is the type of online system identification where the model learns hidden system parameters as a byproduct of optimization itself, eliminating the need for calibration experiments. In this project, we use this approach to learn quadrupole misalignments which are small transverse offsets that effect the beam steering but cannot be directly measured during accelerator operation.

2.4 Cheetah Based Differential Simulation Library

Using physics knowledge based simulator as a prior mean function requires two key properties: the simulator must be fast enough to call within each GP evaluation, and the simulator must be differentiable in order to facilitate gradient based hyperparameter optimization. A beam dynamics code called Cheetah [4] was created to satisfy these needs.

Cheetah uses transfer matrix concept to depict beam transport. For linear optics, each accelerator element transforms the beam's phase space coordinates through a 7x7 transfer matrix. The standard 6x6 transfer matrix Ro is embedded within this structure, with the additional dimension enabling representation of effects such as magnet misalignments. When compared to conventional simulation codes, Cheetah achieves speedups of several orders of magnitude through optimized tensor computation and automatic transfer map reduction. Moreover, all operations in Cheetah is implemented entirely on PyTorch, which means that all operations in Cheetah support automatic differentiation. Gradients of beam properties with respect to any element parameter including magnet position (misalignments) and strengths can be computed efficiently. This ability of differentiation is important for our approach, when GP fits its hyperparameter by maximising marginal likelihood, gradients flow through the Cheetah simulation to update the trainable misalignment parameter. Cheetah provides two beam representations.

- **Parameter Beam:** assumes a Gaussian beam and describes it using a seven dimensional mean vector and covariance matrix, enabling really fast simulation.
- **Particle Beam:** tracks individual macro-particles for higher fidelity results but then there is a tradeoff of higher computational cost.

Since the ARES experimental area runs in the linear regime, where moment-based tracking is sufficiently accurate, this project work uses Parameter Beam.

2.5 The ARES accelerator

The Accelerator Research Experiment at SINBAD (ARES) facility at DESY serves as a testbed for advanced accelerator concepts and to test machine learning techniques on them. It is a normal-conducting S-band linac which produces electron beams up to 155 MeV and is designed for flexibility and reproducibility in research applications.

The project work focuses on the ARES experimental area, as demonstrated in figure 2.2. The beamline consists of three quadrupole magnets (Q1, Q2, Q3) and two corrector/steering magnets (CV for vertical steering and CH for horizontal steering), and a diagnostic screen where beam properties are measured.

The beam tuning task involves optimizing these five magnets:

$$\mathbf{u} = (k_{Q1}, k_{Q2}, k_{Q3}, \theta_{CV}, \theta_{CH}) \tag{2.6}$$

where $\mathbf{u} \in \mathbb{R}^5$ is the control vector.

In order to produce a focused and centred beam at screen, The measured observables are beam's horizontal and vertical positions (μ_x and μ_y) and sizes (σ_x and σ_y). The objective function used throughout this work is the mean absolute error which combines both centering and focusing into a single metric to minimise

$$MAE = \frac{1}{4}(|\mu_x| + \sigma_x + |\mu_y| + \sigma_y) \tag{2.7}$$

Partial observability one of the major challenge in this problem, the quadrupole magnets are never exactly aligned; the precise beam distribution entering the experimental area is not known. Even with the corrector magnets set to zero, a misaligned quadrupole with transverse offset (Δ_x, Δ_y) partially functions as steering element eventually diverting the beam. These misalignments are not directly measurable during operation.

This Simulation-to-Reality gap shows that even an accurate physics model will not perfectly predict beam behavior unless it counts for the actual misalignments. By treating the six misalignments parameters (horizontal and vertical offsets for each quadrupole) as trainable quantities within the physics-informed prior, our approach addresses this issue and allows the model to adjust to actual accelerator settings using the optimization data.

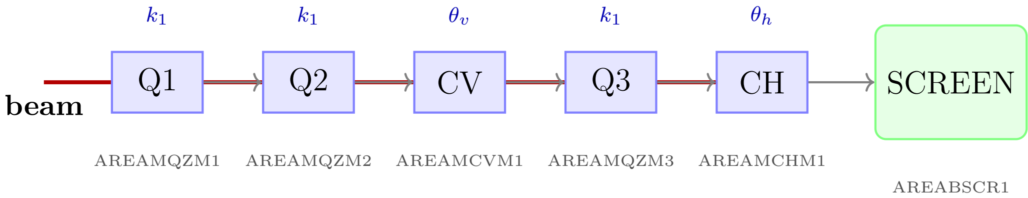

Figure 2.2: Schematic of the ARES experimental area lattice showing the magnet sequence. The beam travels from left to right through quadrupoles (Q1, Q2, Q3) and correctors (CV, CH) before reaching the screen. The adjustable parameters are the quadrupole strengths k_1 and corrector angles θ_v, θ_h.

3 Writing at the Institute of Control Systems

3.1 Hardware

Room N1059 is a computer pool for students to work in. You can use MATLAB/SIMULINK there to do your calculations and write your thesis.

3.2 Software

There are different editors for LaTeX. Some frequently used at the institute are TeXnicCenter and Texmaker. If you use TeXnicCenter you can import the file `Ausgabeprofile\_TeXnicCenter.tcp`, in that folder, by *Ausgabe - Ausgabeprofile definieren... - Importieren*. Choose the output-profile `\LaTeX => PS => PDF student-thesis`. Thus the settings should be correct and a PDF of your work should be generated. If that does not work, check the settings for proper paths to the post-processing and viewer programs.

Visio as graphical software is provided. If you produce figures directly with MATLAB, consider the export-setting of the ”print” command.

3.3 References

The institute uses a common literature database, where a lot of books and articles that you may want to cite are already included. The respective `bib`-file can be found on `S:\ICS library\ics.bib`. Copy the most recent version to your root directory of your thesis!

Then, you can open this file with your LaTeX editor or you can use a literature manager like ”jabref”, for instance. Check if the reference that you want to cite is included. In this case, you can simply use the BiBTeX-key in the respective command. The literature information is properly generated only, if `bibtex` is be executed twice: first `bibtex pd` and then `bibtex main`. If you use TeXnicCenter and the provided output-setting, this is done automatically. In other cases, the easiest way would be to use the `build.cmd` file that accompanies this template. It will execute the compilation process in the proper order and open the pdf file afterwards.

Please provide with your thesis complete information on your references (title, authors, date, DOI, maybe even pdf files). We would like to include these in our database, if this has not already happened..

3.4 Binding the work

The institute offers you to print and bind the work. You need three copies in general, one for your supervisor, one for the Professor and one for yourself, and maybe more. After finalizing your work, it is also included in the literature database by your supervisor.

Your work is printed in the correct order. Put a transparent film in the front and a cardboard in the back.

Your work is stapled. Look for staples of the right length in the shelf in the computer pool and use the large stapler. Afterwards tape the back of your work. You will find illustrative material in the bibliography.

In addition to your printed work, you have to hand in a cd with the pdf file of your work and other important files. Please structure them well, such that some years later somebody else is able to find the important things. The CD is fixed inside your work at the back within a paper cover.

Last but not least, do not forget to sign the declaration.

4 Conclusion and Outlook

This work offers an introduction on how to write a successful bachelor or matser's thesis at the Institute of Control Systems.

Surely, not all your questions have been answered. Ask your supervisor and use the opportunity that he corrects a chapter before handing in your final thesis, to improve your scientific writing!

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